

Solar System

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The Solar System consists of the Sun and objects gravitationally bound to it, including eight planets, dwarf planets, moons, asteroids, and comets.

The Sun is a star composed mainly of hydrogen and helium. Nuclear fusion in its core supplies the energy that reaches Earth as sunlight.

Mercury, Venus, Earth, and Mars are terrestrial planets with solid surfaces. Jupiter and Saturn are gas giants; Uranus and Neptune are ice giants.

Mercury is the smallest planet and closest to the Sun. Venus has a thick carbon dioxide atmosphere and a strong greenhouse effect.

Earth has liquid surface water and an atmosphere dominated by nitrogen and oxygen. Mars has a thin atmosphere and evidence of ancient flowing water.

Jupiter is the largest planet. Its Great Red Spot is a long-lived atmospheric storm. Jupiter's moon Europa has evidence of a subsurface ocean.

Saturn's prominent rings consist mainly of ice particles. Its moon Titan has a dense atmosphere and lakes of liquid hydrocarbons.

Uranus rotates with its axis tilted strongly relative to its orbit. Neptune has powerful winds and a large moon named Triton.

The asteroid belt lies mainly between Mars and Jupiter. The Kuiper Belt lies beyond Neptune and contains icy bodies including Pluto.

Comets contain ice and dust. When a comet approaches the Sun, heating can produce a coma and tails. One astronomical unit is approximately the mean Earth-Sun distance.